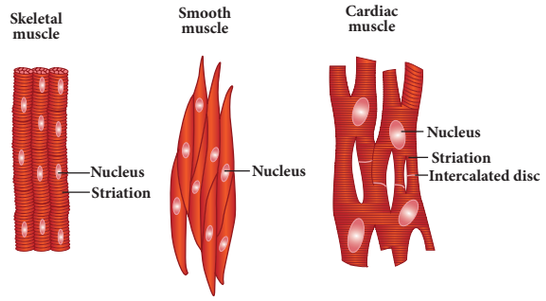


Figure 18.19 Muscle tissue

18.2.4 Nervous Tissue

Nervous tissue comprises of the nerve cells or neurons. They are the longest cells of the body. **Neurons** are the structural and functional units of the nervous tissue. The elongated and slender processes of the neurons are the nerve fibres. Each neuron consists of a **cell body** or **cyton** with nucleus and cytoplasm. The **dendrons** are short and highly branched protoplasmic processes of cyton. The **axon** is a single, long fibre like process that develops from the cyton and ends up with fine terminal branches.

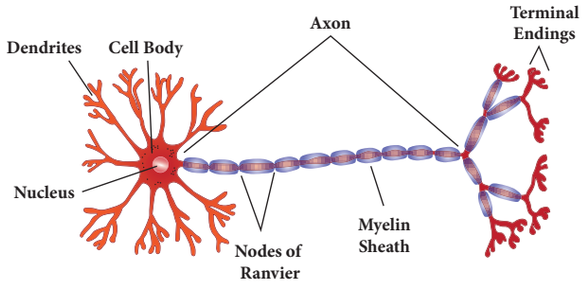


Figure 18.20 Neuron

INFO BIT

Age of our body cells

- Cells of the eye lens, nerve cells of cerebral cortex and most muscle cells last a life time but once dead are not replaced.
- Epithelial cells lining the gut last only about 5 days.

Duration of cell replacement

- Skin cells- about every 2 weeks.
- Bone cells- about every 10 years.
- Liver cells- about every 300 – 500 days.
- Red blood cells last for about 120 days and are replaced.

They have the ability to receive stimuli from within or outside the body and send signals to different parts of the body.



Nerve cells do not undergo cell division due to the absence of centrioles, but they are developed from glial cells by neurogenesis

18.3 Cell Division

Are you aware that all living organisms start their life from a single cell? You may wonder how a single cell then goes to form such a large organism. All cells reproduce by division and division of cells into daughter cells is called cell division.

18.3.1 Types of Cell Division

The three types of cell division that occur in animal cells are:

I. Amitosis - Direct Division
II. Mitosis - Indirect Division
III. Meiosis - Reduction Division

I. Amitosis

It is the simplest mode of cell division and it occurs in unicellular animals, ageing cells and in foetal membranes. During amitosis, nucleus elongates first, and a constriction appears in it which deepens and divides the nucleus into two. Followed by this cytoplasm divides resulting in the formation of two daughter cells.

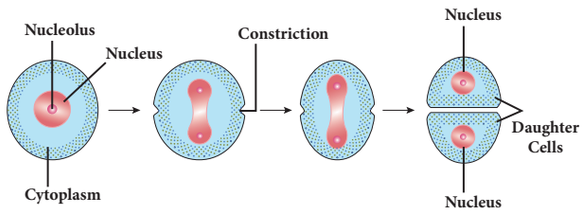


Figure 18.21 Amitosis

II. Mitosis

It was first discovered by Fleming in 1879. In this cell division one parent cell divides into two identical daughter cells, each with a nucleus having the same amount of DNA,

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